1 FROZEN REPRESENTATION

x

raw flow record

f_θ

encoder

parameters frozen

$h(x) \in \mathbb{R}^{128}$

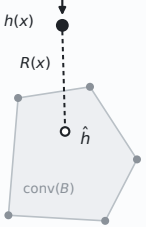
latent code

same checkpoint as the baseline; weights are never updated

2 SCORING OBJECT (same $h(x)$, different distance set)

BASELINE

prototype object



$\hat{h}$ moves with x

$w(x) = \text{softmax}(\phi_\theta(x))$

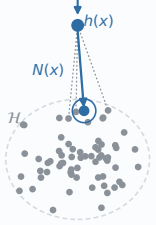
$\hat{h} = B^\top w(x)$

$R(x) = \| h(x) - \hat{h} \|^2$

$B \in \mathbb{R}^{K \times 128}$, $K = 5$ basis vectors

THIS WORK

sample object



$\mathcal{H}$ fixed after training

$\mathcal{H} = \{h_i\}_{i=1}^{n}$

$N(x) = \min_j \| h(x) - h_j \|^2$

$k = 1$ (1-NN)

training normal codes

$s(x)$

3 CALIBRATION

calibration normals

held out from training

$\tau = F_s^{-1}(0.99)$

upper order statistic of $s(x)$

for $s = R$ or $s = N$

$s(x) > \tau$

flag as anomaly

one rule, both scoring objects

Only the scoring object $s(x)$ is varied; the encoder f_θ, the calibration normals and the 99% rule are shared (τ recomputed per s).